

EMCH 260

SOLID MECHANICS

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Mechanical Properties of Materials

3

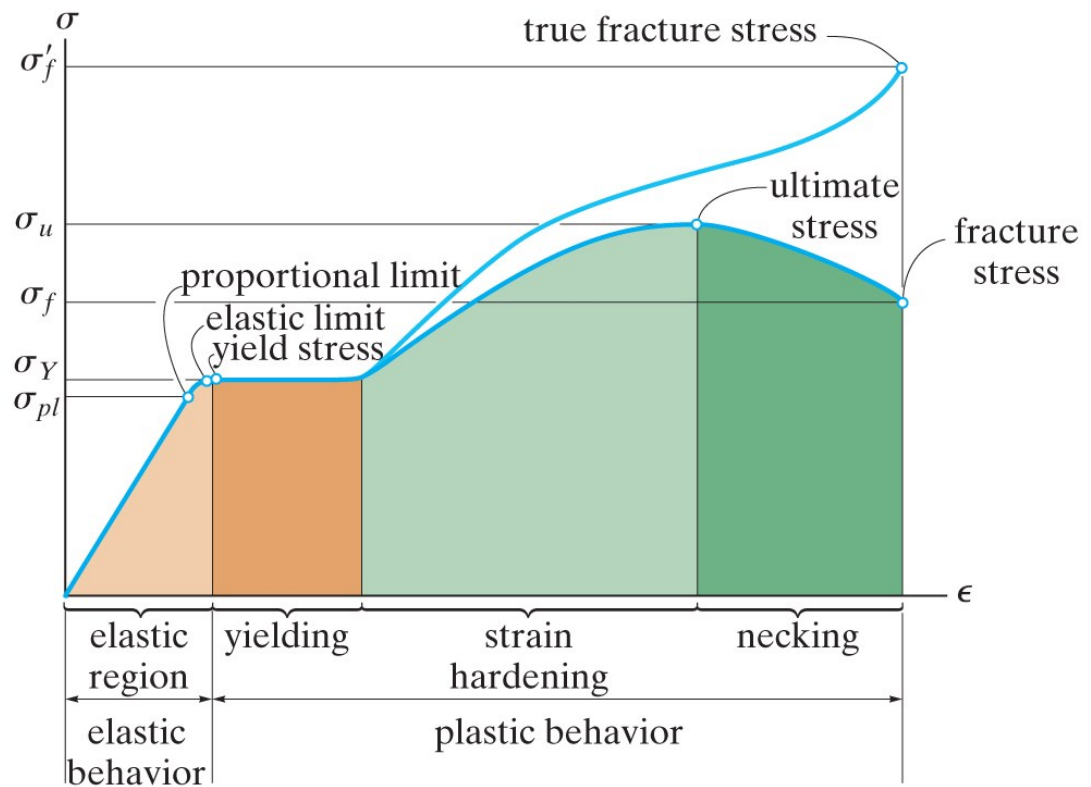
Chapter 3.3

- ✓ Understand shear stress-strain diagram and shear modulus of elasticity
- ✓ Understand and distinguish stress-strain behavior of ductile and brittle materials
- ✓ Understand plasticity and yield strength, know how to use offset method to determine yield strength
- ✓ Understand permanent set, distinguish and determine the elastic and plastic deformation (strains)

CONVENTIONAL STRESS-STRAIN DIAGRAM

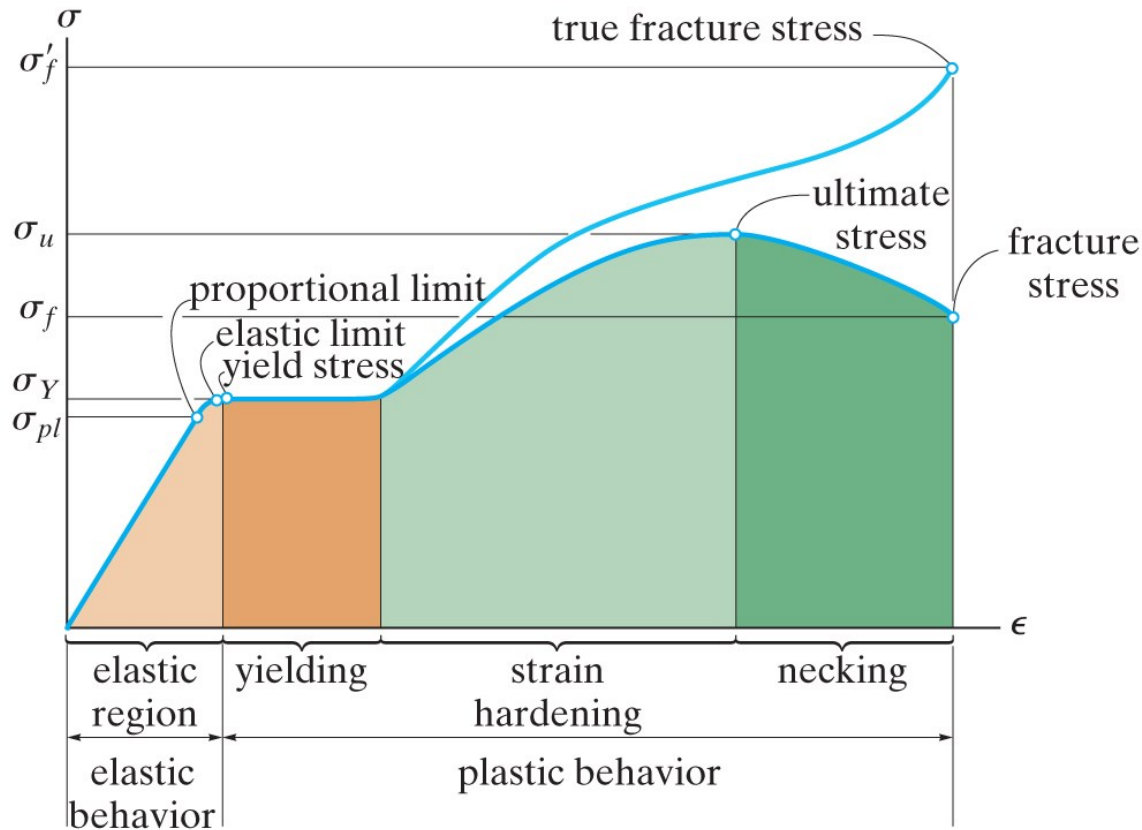
- Note the critical status for strength specification

- Proportional limit
- Elastic limit
- Yield stress (σ_Y)
- Ultimate stress
- Fracture stress

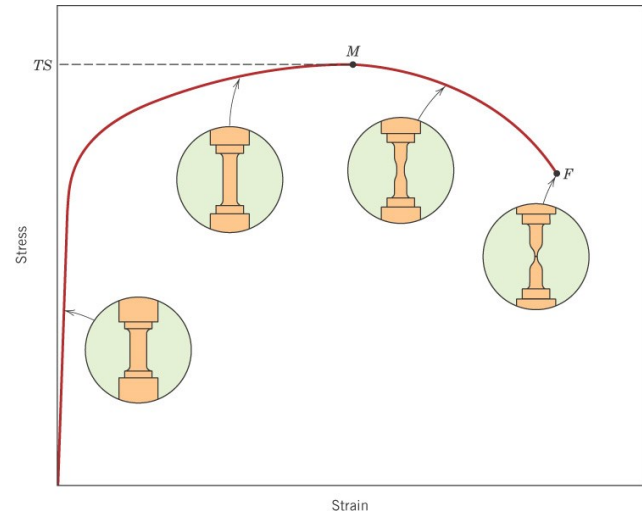


Conventional and true stress-strain diagrams for ductile material (steel) (not to scale)

ULTIMATE STRESS & FRACTURE STRESS



Conventional and true stress-strain diagrams for ductile material (steel) (not to scale)



This steel specimen clearly shows the necking that occurred just before the specimen failed. This resulted in the formation of a “cup-cone” shape at the fracture location, which is characteristic of ductile materials.

DUCTILE MATERIAL

- **Ductile materials** can be subjected to large strains before fracture

- **Percent elongation** =
$$\frac{L_f - L_0}{L_0} \times 100\%$$
$$= \varepsilon_f \times 100\%$$

L_f : length at fracture

L_0 : original length

ε_f : strain at fracture

The *ductility*, in percent elongation, is the plastic strain at fracture.

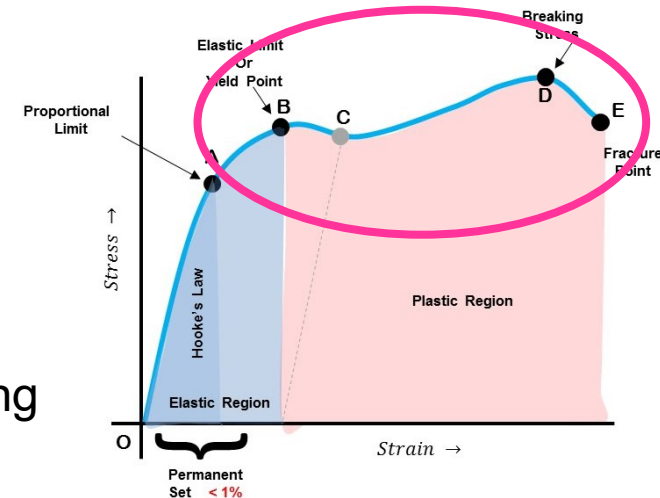
- **Percent reduction of area** =
$$\frac{A_0 - A_f}{A_0} \times 100\%$$

A_f : area of the neck at fracture

A_0 : original cross-sectional area

- **Features of ductile materials:**

- ✓ Elastic behavior
- ✓ Yielding
- ✓ Severe plastic deformation, corresponding to large percent elongation
- ✓ Obvious necking before fracture, corresponding to large percent reduction of area

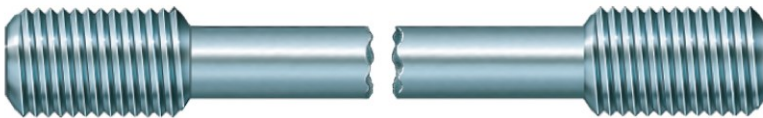


BRITTLE MATERIAL

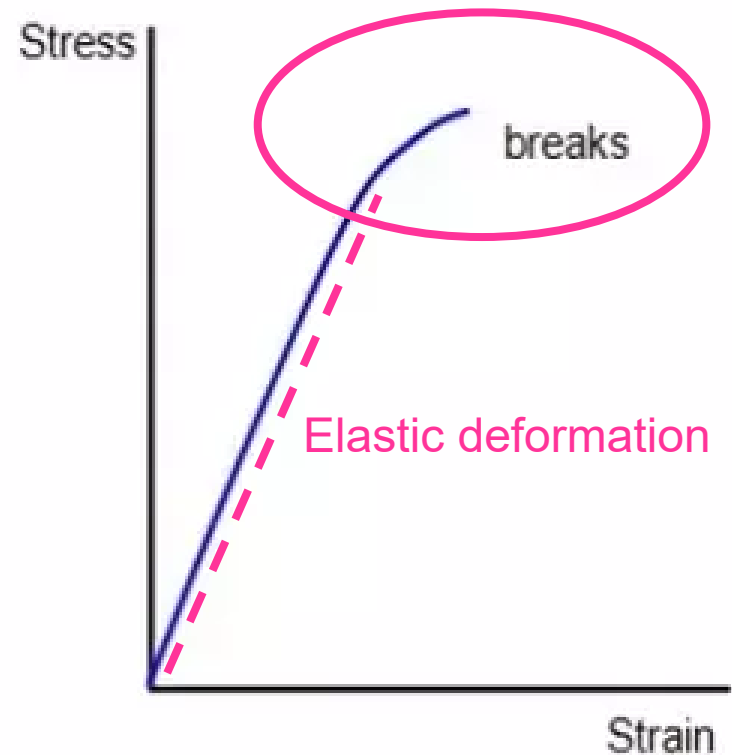
- **Brittle materials** exhibit little or no yielding before failure

- **Features of brittle materials:**

- ✓ Little or no yielding
- ✓ Small or no plastic deformation, corresponding to small percent elongation
- ✓ No necking before fracture, corresponding to nearly zero percent reduction of area
- ✓ Fracture suddenly

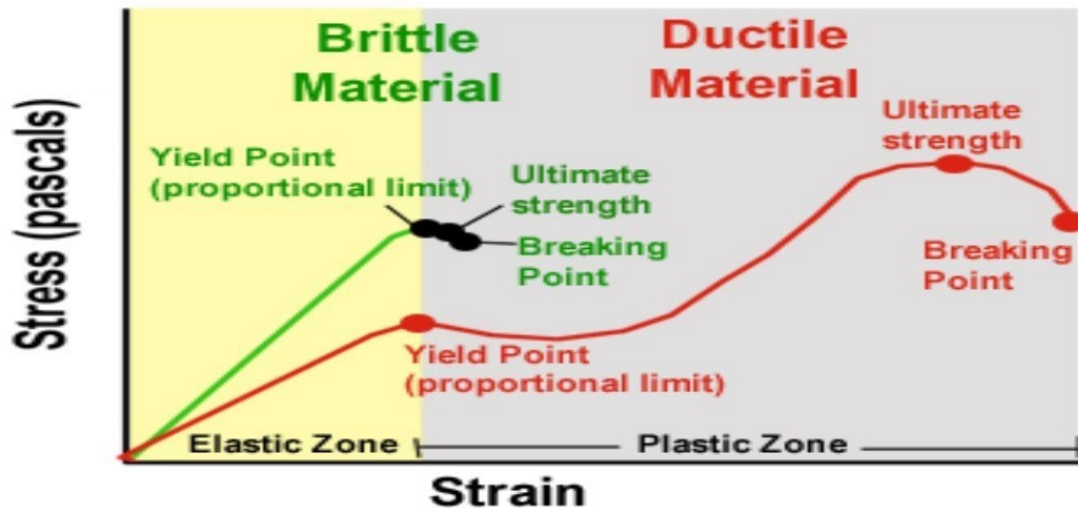


Tension failure of
a brittle material



COMPARISON & EXAMPLES OF DUCTILE & BRITTLE MATERIALS

DUCTILE & BRITTLE MATERIALS



<https://www.youtube.com/watch?v=RGnPq4zTowA>

- **Ductile materials:**

- ✓ Most of metals and some alloys: copper, mild steel (low C%), aluminum, gold
- ✓ Some plastics: nylon ropes, polyethylene

- **Brittle materials:**

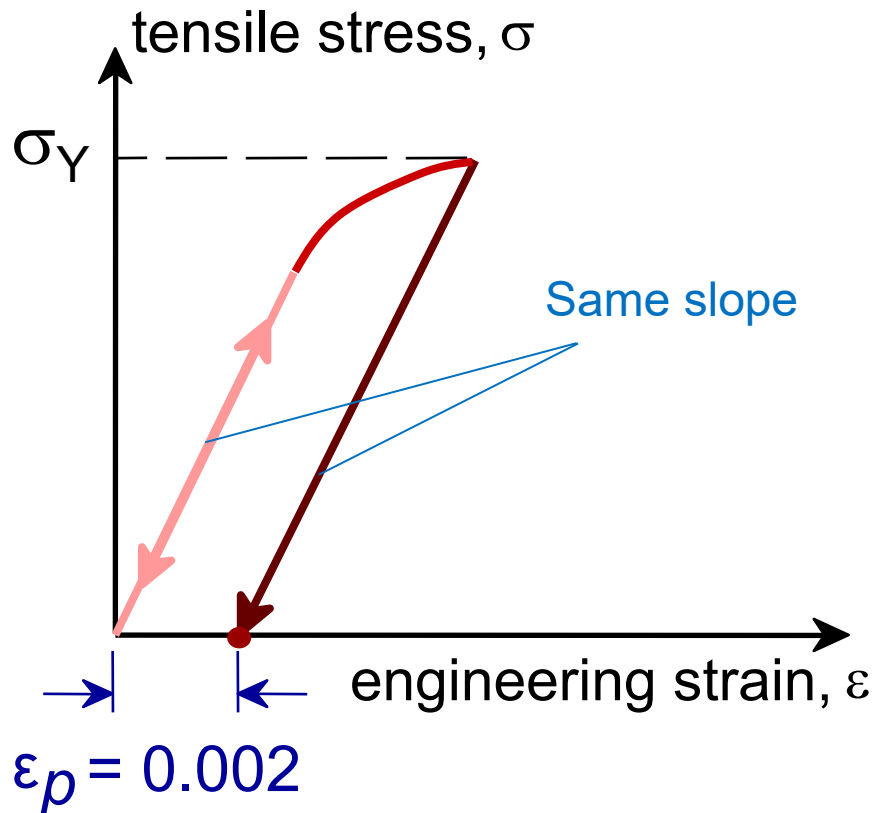
- ✓ Most of non-metal amorphous or inhomogeneous materials: glass, wood, paper, clay, concrete, rocks
- ✓ Some high C% steel

PLASTICITY & YIELD STRENGTH

when $\epsilon_p = 0.002$
or 0.2%

Offset method

- Stress at which *noticeable* plastic deformation (permanent deformation) has occurred.



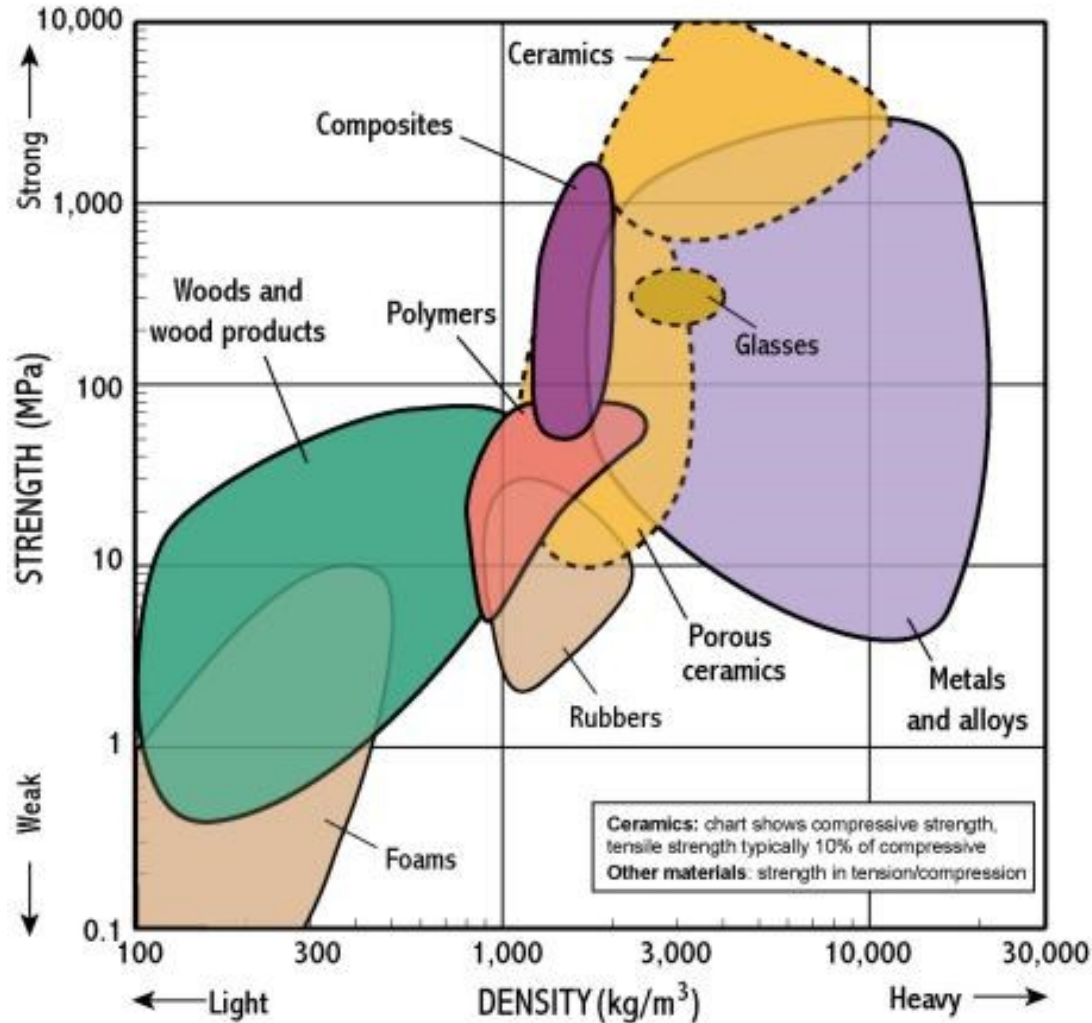
σ_Y : **Yield Strength**

Example: for 2 inch bar

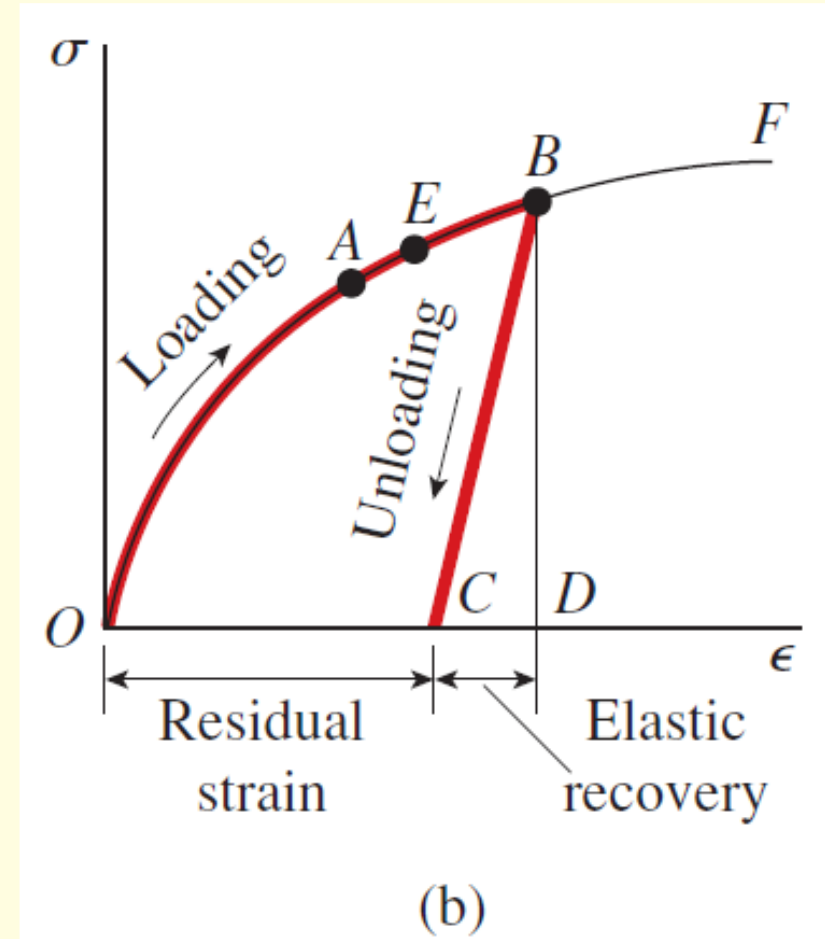
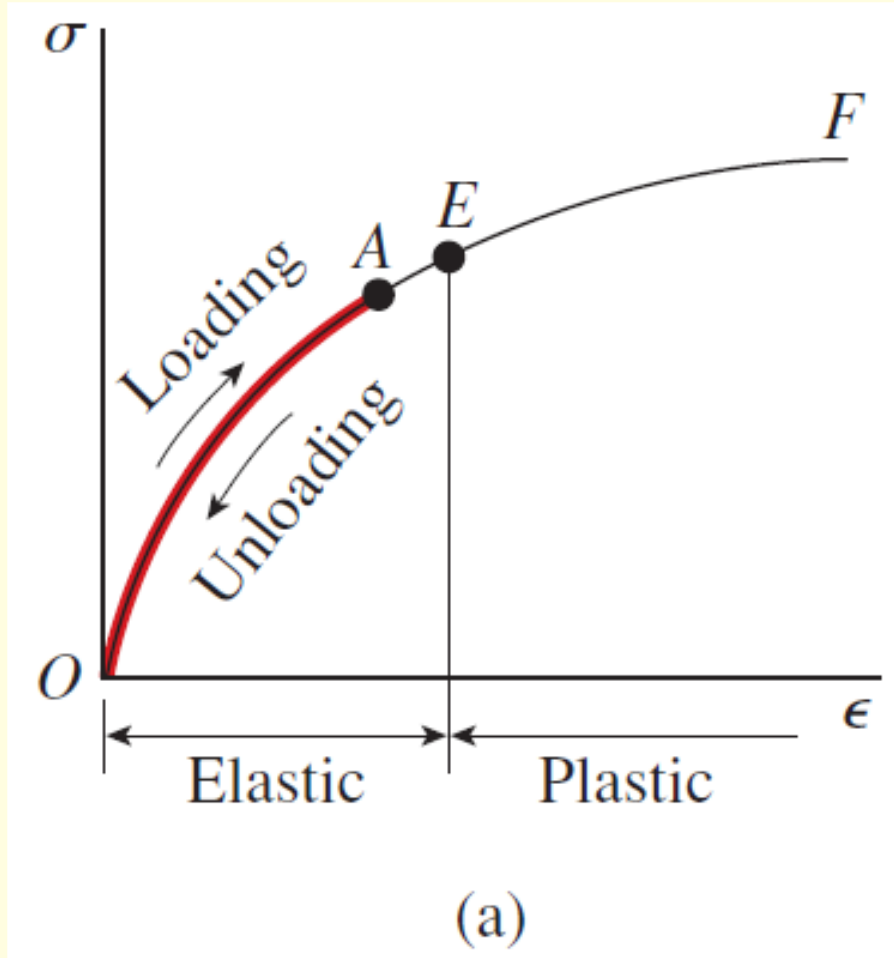
$$\epsilon = 0.002 = \Delta z / z$$

$$\rightarrow \Delta z = 0.004 \text{ in.}$$

MATERIALS STRENGTH



ELASTIC VS. PLASTIC DEFORMATION

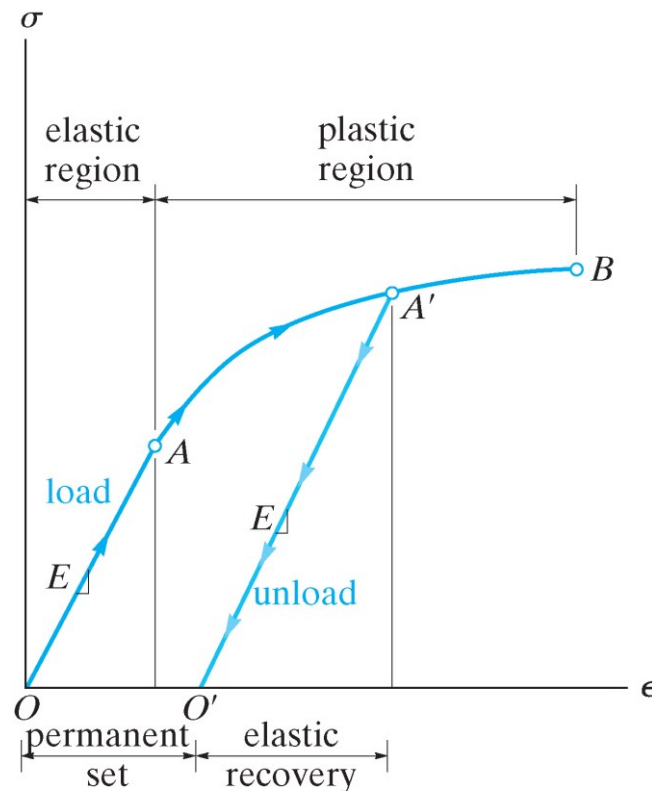


$$\epsilon_{total} = \epsilon_{elastic} + \epsilon_{plastic}$$

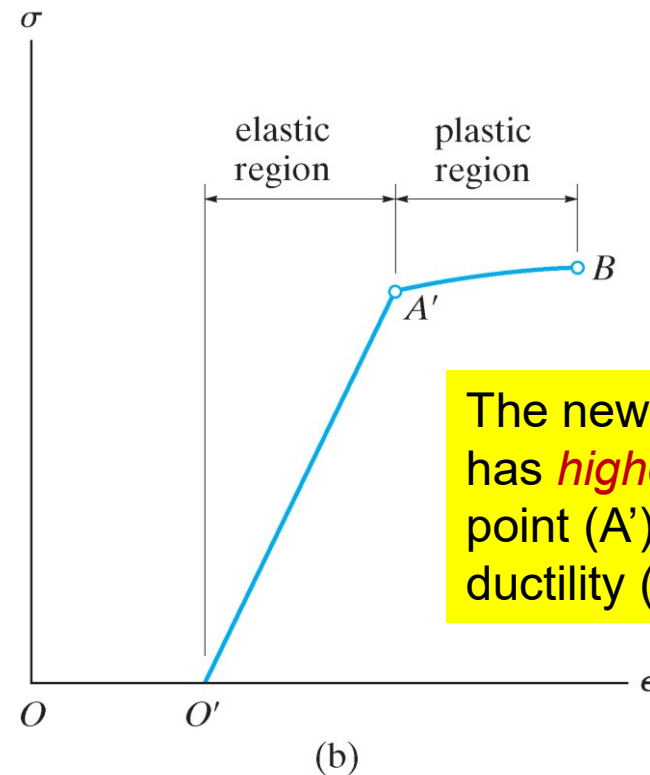
Stress-strain diagrams illustrating (a) elastic behavior, and (b) partially elastic behavior

STRAIN HARDENING

- Unloading from plastic region results in a **permanent set** [OO' in Fig.(a)], although the elastic part can be recovered.
- Reloading from previous permanent set will continue along the same path as previous unloading until the stress reaches the original unloading point [OA' in Fig.(b)], and then the stress continues as before.



(a)



The new diagram has **higher** yield point (A') but **less** ductility ($A'B$)

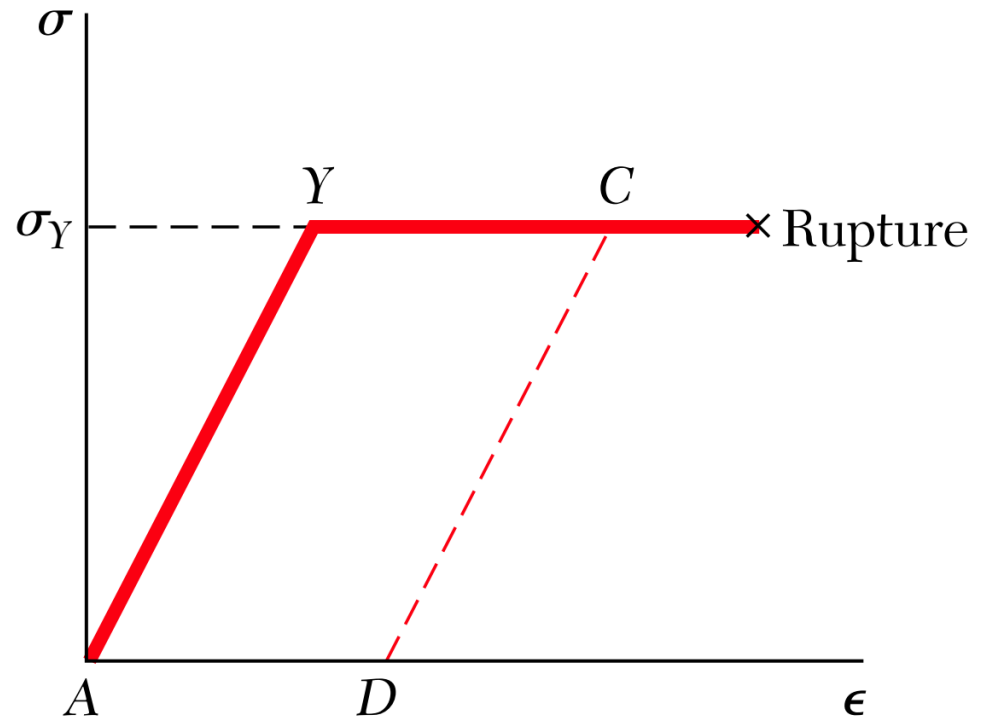
(b)

IDEALIZED ELASTOPLASTIC MATERIAL

- The stress-strain diagram consists of the two straight-line segments

- When σ reaches the value σ_Y , the material starts yielding and keeps deforming plastically under a constant load.

- If the load is removed, unloading takes place along a straight-line segment CD parallel to the initial portion AY of the loading curve.



IN-CLASS EXAMPLE

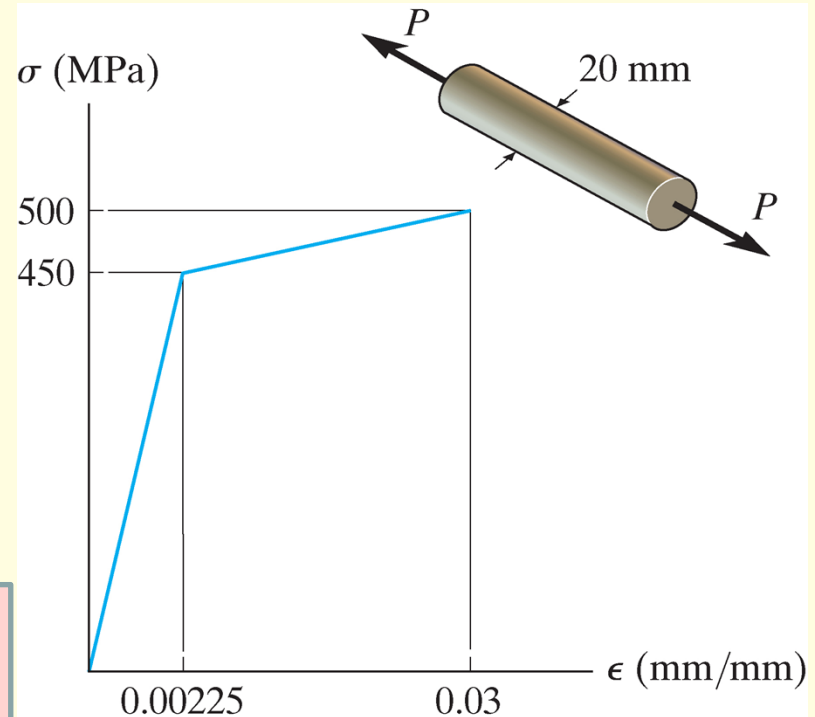
The material for the 50-mm-long specimen has the stress-strain diagram shown. If $P = 157$ kN is applied and then released, determine the permanent elongation of the specimen.

Logic:

Permanent set / deformation / elongation

$$\leftrightarrow \sigma \geq \sigma_Y?$$

$\leftrightarrow$ Stress-strain diagram



IN-CLASS EXAMPLE (cont'd)

Solutions

- The stress in the specimen:

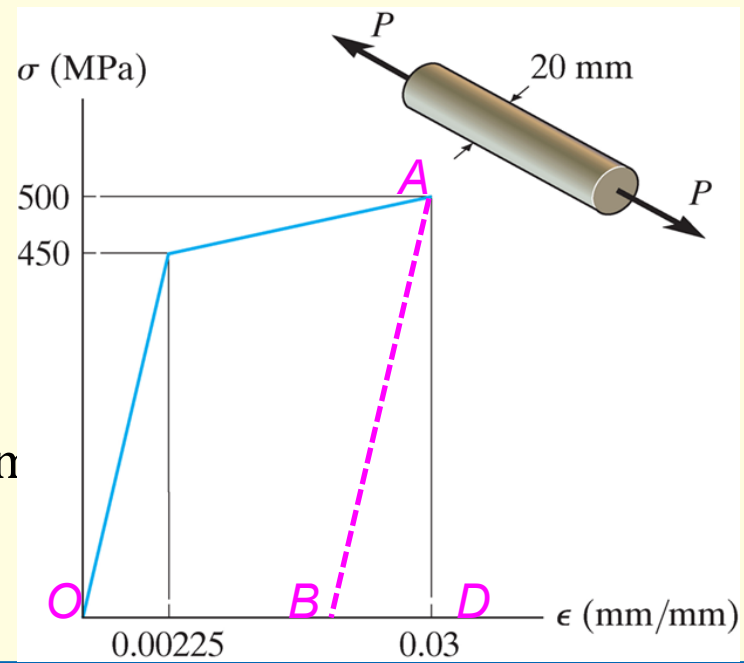
$$\sigma = \frac{P}{A} = \frac{P}{\frac{\pi}{4}d^2} = \frac{157000}{\frac{3.1415926}{4} \times 0.02^2} = 500 \text{ MPa}$$

- Since $\sigma > \sigma_Y = 450 \text{ MPa}$, Hooke's law is not applicable. From the stress-strain diagram, $\epsilon_{\text{tot}} = 0.03$.
- When the load is released, the strain recovers along a line AB which is parallel to the original elastic line.

$$E = \frac{\sigma_Y}{\epsilon_Y} = \frac{450 \times 10^6}{0.00225} = 200 \text{ GPa}$$

- The elastic recovery (BD part) is

$$\epsilon_{\text{el}} = \frac{\sigma}{E} = \frac{5 \times 10^8}{2 \times 10^{11}} = 0.0025 \text{ mm/mm}$$



IN-CLASS EXAMPLE (cont'd)

Solutions, cont'd

- The permanent (plastic) strain (**OB** part) in the specimen is:

$$\varepsilon_{pl} = \varepsilon_{tot} - \varepsilon_{el} = 0.03 - 0.0025 = 0.0275 \text{ mm/mm}$$

- According to the definition of tensile strain, we get the permanent elongation of the specimen:

$$\delta_p = \varepsilon_{pl}L = 0.0275 \times 50 = 1.375 \text{ mm} \quad \textbf{(Ans.)}$$

RELEVANT PRACTICE

- Assignment 1, problem #7 (go to Blackboard under the “Assignments” folder, **Assignment 1 due: 8:30 AM, Monday, Sept. 13th 2021**)
- Optional Extra Exercise 1, problem #5 (go to Blackboard under the “Assignments” folder, **Assignment 1 due: 8:30 AM, Monday, Sept. 13th 2021**)
- *Submit your solutions directly onto Blackboard*

OUT-OF-CLASS READING

- **Textbook, pp. 91 – 100 & 108 – 110**
 - **In particular, Examples 3.1 – 3.3 (pp. 97 – 100) & 3.5 – 3.6 (pp. 109 – 110)**